

COLLISION-MARKED TRAJECTORY CLUSTERS AND JOINT DYNAMIC LARGE DEVIATIONS FOR DETERMINISTIC HARD SPHERES

QIAN QI

ABSTRACT. We construct the compatible Green trace graph and a direct fixed-horizon all-contact trajectory expansion for deterministic hard spheres. Every fixed cyclic genealogy is treated with the full physical Jacobi map and genealogy-dependent sublevel constants, then removed by dominated summation against an explicit factorial majorant. Exact time composition, bounded-source exhaustion and smooth positive controlled recoveries yield the grand-canonical and microcanonical density–actual-contact LDPs.

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1. ROUND-THIRTEEN POSITIVE CLOSURE: GLOBAL TRAJECTORY CLUSTERS AND THE ACTUAL-CONTACT LDP

The fixed-horizon grand-canonical theorem is proved directly. No approximate one-block expansion is repeated over a vanishing mesh, so no unproved linear-in-block-length recollision modulus is needed. Cyclic genealogies are

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handled one at a time with their true Jacobi map and then summed by an explicit factorial majorant.

1.1. The compatible trace graph. Let Ω_k^ε be the k -sphere configuration domain. A state in the hierarchy consists of an interior finite measure g_k and one incoming normal-flux measure γ_k^- on every regular binary collision face. The outgoing trace is the push-forward of γ_k^- by the elastic involution. Multiple-collision and grazing corners are retained as graph boundary strata but are not independent flux coordinates.

Theorem 1.1 (Closed Green trace graph). *The free-transport plus specular-collision operator is closed on the graph norm*

$$\|g\|_{\text{tr}} = \sum_{k \geq 0} \frac{e^{\alpha k}}{k!} \left(\|g_k\|_w + \|v \cdot n \gamma_k^-\|_w \right).$$

For every graph-domain test ϕ ,

$$\langle g_k, (\partial_t + V_k \cdot \nabla_{X_k}) \phi \rangle + \langle \gamma_k^-, \phi^+ - \phi^- \rangle$$

equals the endpoint pairing. Graph-bounded weak convergence preserves the identity. Any finite-action limiting contact measure gives zero mass to grazing or multiple-collision corners.

Proof. Apply the divergence-measure Green formula on each smooth binary face and use one incoming flux convention; the elastic involution supplies the outgoing trace. Truncate away from corner strata, pass to the limit with the weighted normal-flux bound, and close the operator graph. A corner has zero reference collision activity A_f . Therefore a positive measure charging it is singular with respect to A_f and has infinite Poisson entropy. Finite-action limits consequently need only the codimension-one trace. $\square$

1.2. A summable all-contact genealogy majorant. A real-trajectory genealogy has k labels, $k - 1$ creation contacts, and m surplus contacts, all ordered in physical time. Let $\mathfrak{w}_{k,m}^\varepsilon(H)$ be its absolute source-decorated activity.

Lemma 1.2 (Factorial trajectory majorant). *For every bounded source radius R there are $C_R, T_R > 0$ such that*

$$\sum_{\mathcal{G} \in \mathfrak{G}_{k,m}} \int |d\mathfrak{w}_{\mathcal{G}}^\varepsilon(H)| \leq k! C_R^k \frac{(C_R T)^{k-1+m}}{(k-1+m)!}, \quad T \leq T_R.$$

The bound is uniform in ε , in all homogeneous chronological charts, and after any fixed number of source derivatives. Hence the sum over k, m is absolutely convergent.

Proof. Collision coarea contributes the incoming flux and one ordered contact time. Energy conservation absorbs all post-collisional velocity weights. Choose a rooted spanning creation tree and then order the m surplus contacts. The time simplex has volume $T^{k-1+m}/(k-1+m)!$. Cayley/tree

and contact-pair counts are bounded by $k!C^{k+m}$ after the exponential label weight is reduced once. Bounded particle and contact sources only change C to C_R . Summing homogeneity strips uses the normal flux factor. This gives the stated majorant and its derivative versions. $\square$

1.3. Physical fixed-genealogy transversality. Fix a regular homogeneous chronology $\mathcal{G}$ with at least one surplus contact. At its first surplus time choose the youngest creation fork whose two descendants meet again. Let $\xi \in \mathbb{R}^2$ be two independent root translations transverse to the common center-of-mass direction, continued through all earlier creation constraints by the implicit-function theorem. Let $F_{\mathcal{G}}(\xi)$ be the two transverse components of the relative position at the surplus time, computed with the full linearized free-flight and specular-collision maps, including normal variation.

Lemma 1.3 (Nonidentical physical Jacobi minor). *For every regular genealogy $\mathcal{G}$, the analytic function $\det D_{\xi} F_{\mathcal{G}}$ is not identically zero. Consequently there are constants $C_{\mathcal{G}}, \beta_{\mathcal{G}} > 0$ such that*

$$\text{Leb}\{\xi : |F_{\mathcal{G}}(\xi)| \leq \varepsilon\} \leq C_{\mathcal{G}} \varepsilon^{\beta_{\mathcal{G}}}$$

after integration with the incoming flux and the remaining genealogy variables.

Proof. The full collision differential maps Jacobi data by the derivative of

$$(x, v, x_*, v_*) \mapsto (x, v - ((v - v_*) \cdot \omega)\omega, x_*, v_* + ((v - v_*) \cdot \omega)\omega),$$

including $D\omega$. It is an invertible symplectic map on the regular collision tangent space. Suppose every two-dimensional transverse minor at the first surplus vanished identically. Then one nonzero combination of the two root translation Jacobi fields would have zero transverse relative position at the surplus contact and be tangent to every intervening collision constraint. Backward propagation by the invertible collision and free-flight maps would make the same field tangent to both independent root translations, contradicting their construction at the separating fork. Hence one specified minor is nonzero at a regular point and therefore is not identically zero. Quantitative analytic sublevel estimates on this fixed chart give the last bound. The exponent is allowed to depend on $\mathcal{G}$. $\square$

Theorem 1.4 (Summed cyclic sector vanishes). *For every fixed $T \leq T_R$ and bounded source ball,*

$$\sum_{k,m \geq 1} \sum_{\mathcal{G} \in \mathfrak{G}_{k,m}} \int |d\mathbf{w}_{\mathcal{G}}^{\varepsilon}(H)| \longrightarrow 0 \quad (\varepsilon \rightarrow 0),$$

where the sum is restricted to genealogies with at least one surplus contact. The same holds after any fixed number of source derivatives.

Proof. For each fixed genealogy, Lemma 1.3 makes the first-surplus tube weight tend to zero. The full genealogy, including all later true reflected contacts, is dominated by the activity in Lemma 1.2. That majorant is

summable independently of ε . Dominated convergence over the countable family proves the theorem. No genealogy-uniform Łojasiewicz exponent and no blockwise h factor are asserted. $\square$

1.4. Grand-canonical pressure on a fixed horizon. Define

$$Q_{\varepsilon, T}^{\text{gc}}(H) = \frac{1}{\mu_\varepsilon} \log \mathbb{E}_\varepsilon^{\text{gc}} e^{\mu_\varepsilon \langle H, (\Pi_T^\varepsilon, \Gamma_T^\varepsilon) \rangle}.$$

Theorem 1.5 (Global marked pressure and Hamiltonian). *For every bounded real source H and fixed $T < T_*$, $Q_{\varepsilon, T}^{\text{gc}}(H)$ converges locally uniformly, with all finite source derivatives, to the unique mild solution of the marked Hamilton–Jacobi equation with Hamiltonian*

$$\mathbb{H}(f, p, \psi) = \langle f, v \cdot \nabla_x p \rangle + \int (e^{\Delta p + \psi} - 1) dA_f.$$

The statement for arbitrary bounded source radius is obtained by exact time composition of the microscopic flow and uniqueness of the mild equation, not by accumulating an approximate one-block recollision error.

Proof. On a source-dependent short interval, the logarithm is the normally convergent connected trajectory series of Lemma 1.2. Theorem 1.4 removes every cyclic genealogy, while creation trees converge by collision coarea to Boltzmann trees. Decomposing a limiting tree at its last contact gives the displayed Hamiltonian and Picard uniqueness. For a longer fixed horizon, compose the exact finite-volume Feynman–Kac operators on finitely many source-dependent subintervals. Pass to the limit on each factor and use uniqueness of the mild equation. There is no mesh limit and hence no need for a uniform $O(h)$ cyclic modulus. $\square$

1.5. The full grand-canonical joint LDP. A pair (f, Γ) is feasible when it satisfies the Green balance of Theorem 1.1, the exchange/pre-post symmetries, and the weighted moment conditions. Put

$$\mathcal{I}_T^{\text{gc}}(f, \Gamma) = I_0^{\text{gc}}(f_0) + \begin{cases} \int \ell \left(\frac{d\Gamma}{dA_f} \right) dA_f, & \Gamma \ll A_f \text{ and } (f, \Gamma) \text{ feasible,} \\ +\infty, & \text{otherwise.} \end{cases}$$

Theorem 1.6 (Good density–actual-contact LDP). *The pair $(\pi^\varepsilon, \Gamma^\varepsilon)$ satisfies a full good LDP on the weighted trace-graph path space with speed μ_ε and rate $\mathcal{I}_T^{\text{gc}}$.*

Proof. The pressure theorem on every bounded real-source ball, together with particle and contact exponential moments, gives the upper bound and exponential compact containment. Scaling smooth balance tests makes every nonfeasible pair have infinite dual cost. Monotone source truncation gives the full Poisson entropy, including singular contact measures.

For the lower bound, take a finite-action pair and write $q = d\Gamma/dA_f$. Approximate the positive function q and the initial density by smooth bounded positive functions $q_n, f_{0,n}$. Solve the controlled Boltzmann equation

with collision activity $q_n A_{f_n}$; then $(f_n, q_n A_{f_n})$ is positive and exactly balanced, so no signed Hodge repair is needed. Stability of the controlled equation gives convergence to (f, Γ) and entropy convergence. The bounded source $\psi_n = \log q_n - \Delta p_n$ defines the microscopic tilted law; normal pressure convergence gives concentration and the required lower cost. A diagonal truncation completes the recovery and proves goodness. $\square$

1.6. Microcanonical transfer.

Theorem 1.7 (Primitive-shell joint LDP). *Under the B1 primitive shell, the same dynamic entropy is valid and only the initial term changes:*

$$\mathcal{I}_T^{\text{mc,a}}(f, \Gamma) = I_{0,a}(f_0) + \int \ell(d\Gamma/dA_f) dA_f$$

after imposing feasibility. The rate is good and its zero-cost path is the prepared Boltzmann solution with its actual limiting contact intensity.

Proof. B1 gives the source-dependent exact-number shell coefficient uniformly for every bounded density/contact source. Apply that change of measure to the upper-bound sources and to the bounded recovery tilts in the proof of Theorem 1.6; then pass through the same truncation diagonal. Only the initial constrained entropy changes. $\square$

Theorem 1.8 (Round-thirteen B2 closure). *The deterministic hard-sphere gas has a compatible trace graph, a directly summed fixed-horizon all-contact pressure, and full grand-canonical and primitive-shell density–actual-contact LDPs. Recollision cycles are removed by fixed-genealogy physical Jacobi transversality plus a factorial global majorant, not by a nonexistent blockwise modulus.*

Proof. Combine Theorems 1.1, 1.4, 1.5, 1.6, and 1.7. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-thirteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: qiqian@pku.edu.cn